MAASTRICHT UNIVERSITY

Norway's Dependency on Oil and Future Developments

Tutorial Group 1

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1. Introduction

The country of Norway with its population of just above five million belongs to the most prosperous countries in the world regarding gross domestic product (GDP) per capita (OECD, 2019). Furthermore, the people in Norway enjoy one of the highest Human Development Indexes, as well as their defining inclusive economy (United Nations Development Programme, 2019). A considerable amount of Norway's current wealth and economic strength, however, can be traced back to the discovery of natural resources, namely oil, and their strong public sector, which ensured that the returns of those resources would be handled in a sustainable matter (Norges Bank, 2019). Nevertheless, having a petroleum sector as large as one fifth of one's small and open economy can bring its own risks and concerns for a country (Ministry of Petroleum and Energy, 2019). Especially given recent times of increasing worldwide uncertainty and diminishing oil returns, the actions of the Norwegian government to reduce their country's dependency on oil become all the more important. This paper aims to illustrate Norway's dependency on oil by applying a theoretical framework and comparing its prediction to empirical data. Following this backward-looking analysis, this paper addresses the efforts and tools of the government, including Norway's social welfare fund, to diversify their economy and put those in context of a future outlook.

Firstly, the paper provides sufficient context of Norway's current economic situation and background, mainly focusing on developments in the oil production and the creation of Norway's social welfare fund. Secondly, starting our analysis, the paper adopts the theoretical framework from Galí (2018) and applies exogenous shocks presenting events outlined in the background. The predictions resulting from the model are then put in perspective with empirical data. Thirdly, an outlook for Norway's economy is constructed by considering both the social welfare fund and oil dependency and linking it to Norway's Economic Outlook, developed by the Organisation for Economic Co-operation and Development (OECD). Lastly, the paper formulates a concluding remark summarising the results of the backward-looking analysis and the forward-looking outlook.

2. Norway's Past Developments

There are an extensive number of factors that determined Norway's position in today's global economic landscape. However, the paper mainly addresses the interplay of Norway's petroleum sector and its whole economy. Therefore, the focus of describing past developments in Norway is divided into two parts. This section, first, gives an overview of oil production in Norway since its discovery in 1969 and second provides an introduction to the Government Pension Fund of Norway, which was established in 1990 from the returns of the petroleum sector.

2.1 Oil Production

Norway's history of oil began a few decades ago in the 1960s, when various private companies were granted licenses by the Norwegian Government to drill for oil. The first major discovery was at the location Ekofisk in 1969, and more followed throughout the 1970s. Norway's public presence in the still young petroleum sector was formed early on in 1972 with the establishment of the state-owned company Statoil (nowadays Equinor) as well as a minimum requirement for the states ownership in production licenses. However, the structure of the government's influence in the sector was restructured through the introduction of the State's Direct Financial Interest (SDFI) in 1985, which represents a portfolio of the state's ownership in various facilities and companies. Nowadays, the government essentially possesses three main sources of income from the petroleum sector; returns on the ownership in Equinor, returns on its SDFI portfolio, as well as tax returns from private companies operating in the sector. Both oil crises in the 1970s were experienced by the Norwegian economy and petroleum sector. Especially, the price drop after the second oil crisis in 1980s strongly affected the economy and is according to the Norges Bank (2004) one of the causes that led to the Norwegian banking crisis in 1987. Furthermore, recent oil price development worth mentioning for the analysis include rapid growth during the years prior the financial crisis in 2008 and a sharp decrease in the oil price in 2014 due to lower global demand.

2.2 Government Pension Fund

In 1990, after producing oil for nearly two decades, the Norwegian government decided to set up a social welfare fund (SWF) as the Petroleum Fund of Norway. In 2006, the name was changed, and the Petroleum Fund of Norway is now known as the Government Pension Fund Global. The primary purpose of this social welfare fund is to protect the Norwegian economy

from fluctuations in the oil price by investing revenues from oil and gas activities into the fund (Norges Bank Investment Management, 2019). This procedure ensures that oil revenues are spent conscientiously and intends to secure Norway's wealth in the long run as well as in the present. By investing the revenues from petroleum activities, the fund grows with increased oil production and with higher returns from previous oil proceeds. Furthermore, the fund only invests into foreign assets, so as to ensure that the strength (or weakness) of the Norwegian economy does not affect the capital income from the fund. Due to the long investment horizon, the Norwegian government only invests into companies whose values are aligned with society's interests. Therefore, it abstains from investing into unethical companies and has already excluded companies from their portfolio, such as 'Boeing Company' or 'Honeywell International Inc.' for being involved with nuclear weapons (Norges Bank Investment Management, 2007). Since 2011, the Government Pension Fund is the largest sovereign welfare fund in the world, having surpassed $1 trillion in assets in September 2017 (The Economist, 2017). To make sure that not all of the available fund money is spent at once, the government is allowed to spend the expected real return of the fund, which is approximated to be 3 percent per year, in order to finance fiscal policies (Norges Bank Investment Management, 2019). This policy ensures that Norway can benefit from oil revenues in the future and the present, even when all oil and gas resources are gone.

3. Oil Dependency

Norway's petroleum sector has grown to a size as large as 24% of Norway's total GDP in its peak in 2006. Since then, the industry has decreased relative to the whole economy with a share of 14% of GDP in 2019 (Ministry of Petroleum and Energy, 2019). Nevertheless, Norway's dependency on oil, or rather on the oil price and worldwide demand, has not vanished in the last years. This is easily observable by plotting the price of oil against Norway's exports or the exchange rate of the Norwegian Krone (NOK) to other currencies. Figure 1 illustrates the comparison between the price of oil and Norway's exports, while Figure 2 analyses the second statement, that the Norwegian exchange rate reacts to and moves with the oil price to some extent.

To fully understand how Norway reacts to shocks caused by oil, this paper focuses its analysis on the price decrease of oil in 2014. Starting from the two abovementioned assumptions, the

paper applies the IS/MP-AD/AS model based on Romer (2000) and Galí (2018). to illustrate the shock and relates that to actual data and reports of that time.

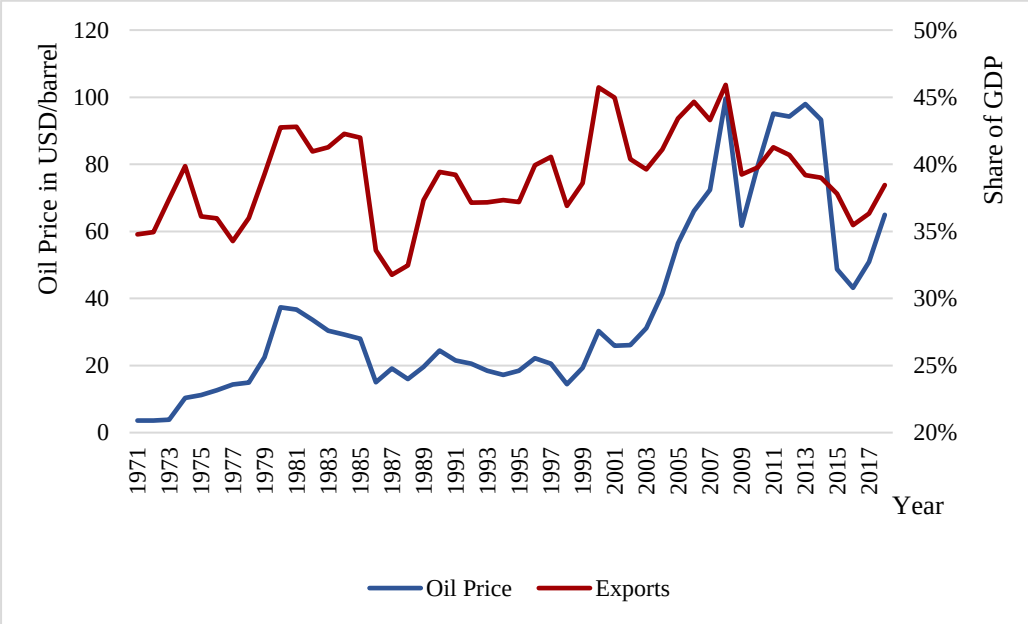

Figure 1: Brent crude oil price and export as a percentage of GDP. Retrieved from Federal Reserve Bank of St. Louis and OECD (2019).

3.1 Oil Price Drop in 2014

During the first decade of this millennium, emerging markets, amongst others China, Russia and Brazil, experienced rapid growth and therefore the demand for oil was increasing. After the financial crisis in 2008, the demand for oil began to drop and economic growth began to slow down. The impact on oil price in 2014 was especially large because many of those emerging markets, particularly China, lowered their demand drastically.

3.1.1. Initial Shock

Empirically, both assumptions that exports decrease, and the NOK depreciates still hold for the price drop in 2014. Referring to Figure 1, it can be observed that total exports decreased by 4% as a share of GDP between 2014 and 2016. In addition, the exports of the petroleum industry alone have decreased by 10 percentage points compared to total exports in those two years. Due to the reduction in oil price, the petroleum sector of Norway became less profitable (see Figure 8 in the appendix). As resources are located offshore, extracting these present a costly procedure (Moses & Letnes, 2017). Thus, any shocks altering the profitability heavily influence the capital invested into that sector. Indeed, investments in the petroleum sector were 199 billion NOK in 2014. In the years following, investments decreased until reaching their lowest point since 2008

with a value of 127 billion NOK (Ministry of Petroleum and Energy, 2019) (see Figure 9 in the appendix).

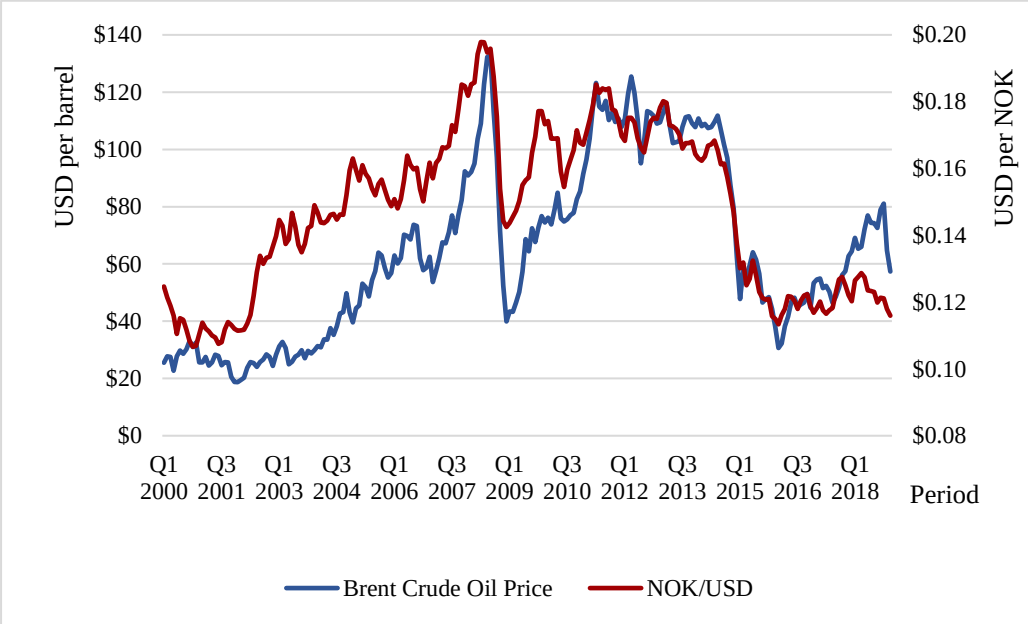

Figure 2: Brent crude oil price and exchange rate NOK/USD. Retrieved from (Federal Reserve Bank of St. Louis, 2018).

Figure 2 depicts the Brent crude oil price in US dollar per barrel and the exchange rate of the US dollar per NOK. The two variables clearly show a common trend from 2010 onward, almost moving jointly in the same direction, confirming our assumption that the NOK depreciated in 2014. However, Norges Bank (2015) stated in their Monetary Review that the depreciation only partially resulted from lower oil prices. Nonetheless, Norway's floating exchange rate functions as a shock absorber for its economy, allowing other industries to export more easily, being cheaper on the worldwide market (Al-mulali, 2010). The initial reduction of oil exports is thus partially offset by an increase in exports of other sectors (Norges Bank , 2015).

In order to illustrate this exogenous shock, the paper applies a basic modified IS/MP-AD/AS model based on Romer (2000) and Galí (2018). The equation for our IS-curve is given by

$$\tilde{y} = \tilde{y}^e - \sigma^{-1}(i - \pi^e - r^n) + \varepsilon_{\tilde{y}}$$

in which $\tilde{y}$ is the output gap (expressed as the difference between log output y and log natural output y^n), i represents the nominal interest rate, π^e denotes the expected inflation rate (here, the target), r^n is the natural rate of interest and where $\varepsilon_{\tilde{y}}$ represents temporary shocks to the demand-side of an economy. As stated above, the weaker economic activity in the petroleum sector lowers investment as well as employment. This, in addition to rising global uncertainty

will potentially reduce private consumption and investment in other industries as well (Norges Bank, 2014). All in all, this represents a negative demand shock, illustrated by a leftward shift of the IS curve in Figure 3.

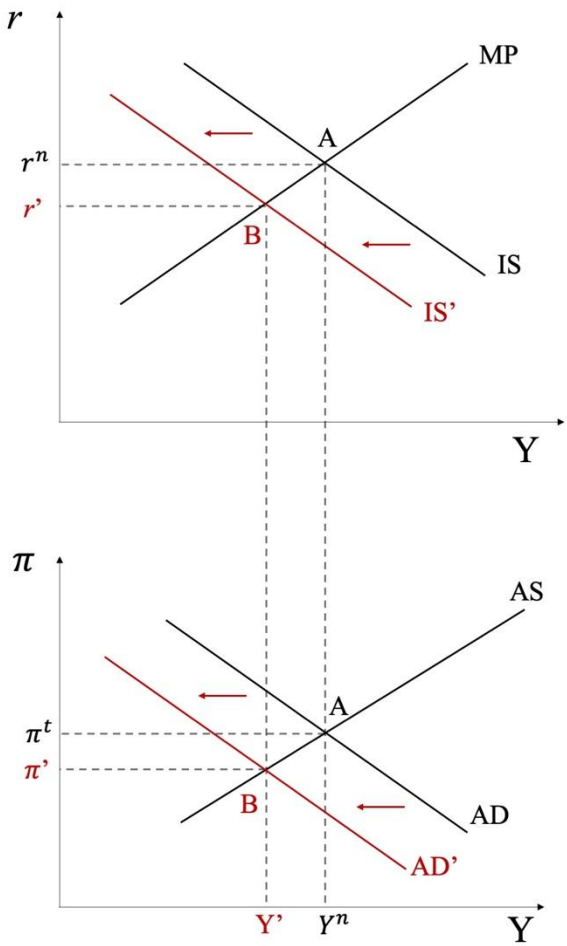

Figure 3: Negative demand shock in IS/MP-AD/AS model. Adapted from Lukkezen J., Jacobs B. & Kool C. (2016) by authors.

This shock will be considered a temporary shock, only changing the $z_{\tilde{y}}$ in the IS equation. In addition, Norges Bank has been proved to be quite credible following its inflation target despite floating exchange rate, which will cause expectations to only adjust slowly away from the target. The AS-curve is based on expected and not previous inflation and hence does not shift. This implies that the economy will be located at point B immediately after shock with a lower interest and inflation rate as well as a negative output gap. In fact, data from OECD (2019) shows how the negative output gap of Norway increased from 2014 onwards as depicted by Figure 4. Moreover, the unemployment rate also started to rise significantly (see Figure 10 in the appendix).

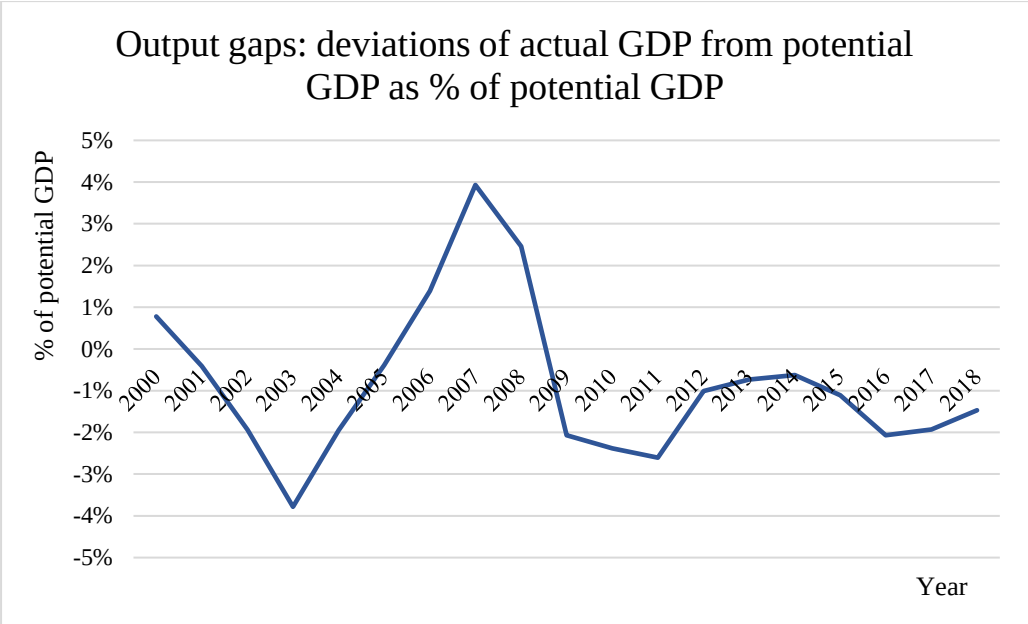

Figure 4:Output gaps: deviations of actual GDP from potential GDP as a percentage of potential GDP. Retrieved from (OECD, 2019).

3.1.2. Monetary Reaction

The reaction of the monetary policy is illustrated by the MP-curve, which is given by the following equation

$$i = i^T + \phi_\pi(\pi - \pi^T) + \phi_{\tilde{y}}\tilde{y} + \varepsilon_i$$

The crucial parameters are ϕ_π and $\phi_{\tilde{y}}$, which represent a central bank's importance to stabilise inflation and output, respectively. Norges Bank follows flexible inflation targeting, putting emphasis on both, stable inflation and maximised capital utilisation (Norges Bank , 2015). Due to the temporary nature of the shift, the MP-curve does not shift. This, however, does not indicate that the central bank ignores the shock. Instead, the response is illustrated in a movement along the MP-curve from A to B. Figure 5 shows the lowering of the policy rate conducted by the Norges Bank throughout 2015. Concluding, the Norge Bank leans against the wind to counteract and partially offset the adverse effect of the initial demand shock.

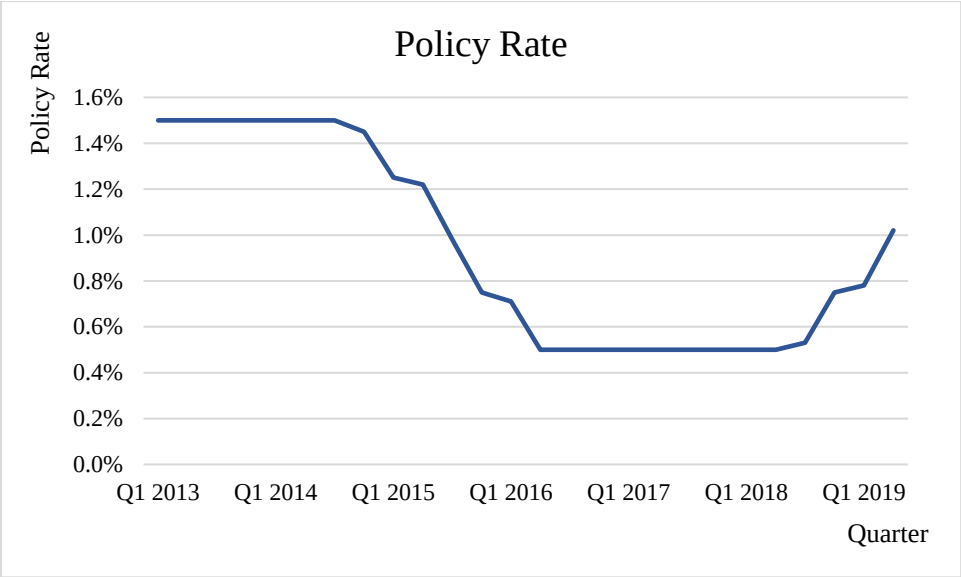

Figure 5: Policy rate as a percentage per year. Retrieved from Norway Ministry of Petroleum and Energy (2019).

3.1.3. Fiscal Reaction

Besides a lower interest rate and negative output gap, Figure 3 also implies lower inflation following the shock. However, Figure 6 illustrates clearly that this has not been the case. Not only that, but inflation actually sharply increased in 2016.

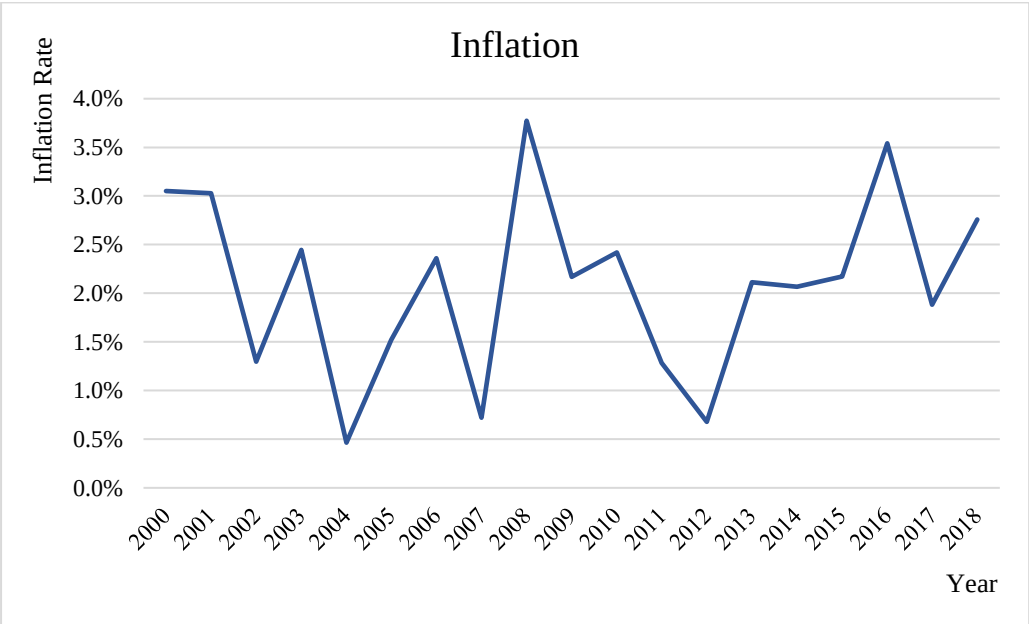

Figure 6: Inflation as a percentage rate. Retrieved from OECD, (2019).

One possible explanation of this contradiction could be through fiscal policy. In the model, the economy would slowly, but eventually return to its natural level of output. In order to accelerate that process, the government could use expansionary fiscal policy to bring its economy back to the equilibrium. From 2014 to 2016, the government decided to decrease the expenditure on

petroleum activities while increasing the expenditures on non-petroleum activities on a larger scale. The total expenditure increased continuously, from 1127.1 billion NOK in 2014 to 1194.5 billion NOK in 2015 up to 1259.1 billion NOK in 2016 (Norwegian Ministry of Finance, 2016).

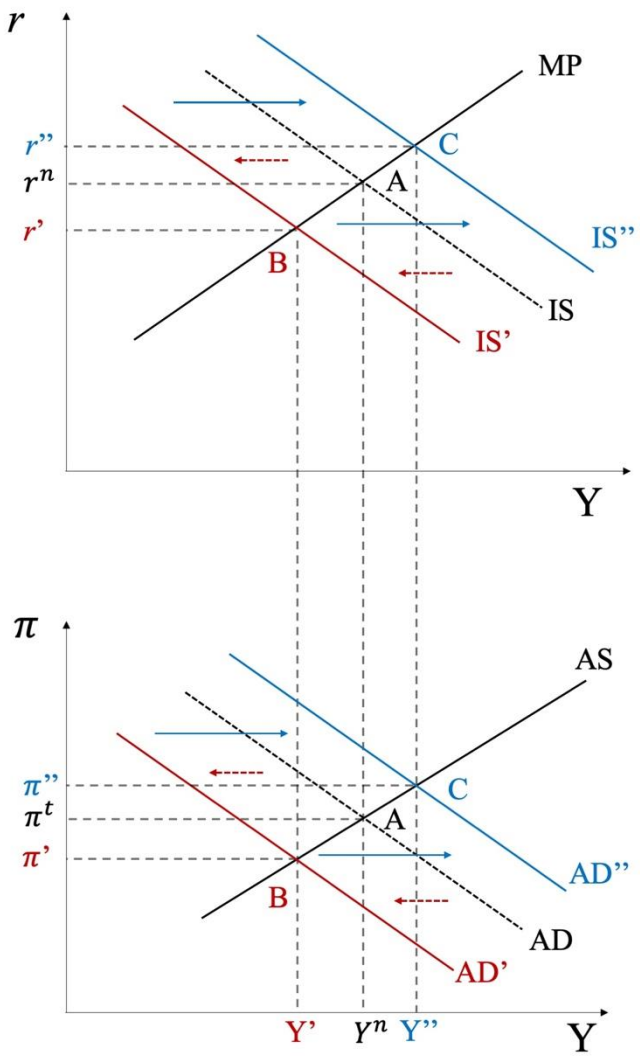

Figure 7: Overshooting effect in IS/MP-AD/AS model. Adapted from Lukkezen J., Jacobs B. & Kool C. (2016) by authors.

For fiscal policy to be the cause of increased inflation, the government would have had to overshoot their expansionary fiscal policy as illustrated by the rightward shift of the IS-curve in Figure 7. However, this would also indicate a positive output gap which was clearly not existent in 2016. Rather, Norway's increase in expenditure most likely offset the adverse shock partially, allowing the economy to return to its natural level faster but not completely. The high inflation in 2016 is most likely not easily explained by the simplified closed economy model used. Norge Bank (2016) reports that the sharp increase in inflation in the beginning of 2016 is

primarily due to open market conditions, namely the depreciation of the NOK. This generates relatively more expensive imports for Norway (goods and raw materials), which raises the cost of production and therefore, the final price paid by consumers as well.

Based on data from OECD (2019) and Norges Bank (2019), Norway supposedly returned to its natural rate of output in the beginning of 2019. Norges Bank's decision to lower its inflation target from 2.5% to 2% in 2018 most likely plays a crucial role in reaching that point as well. Relating to the model, this implies a downward shift of the MP and AS-curve, since expected inflation is going to decrease. Figure 8 depicts the return to the natural output combining the aforementioned reactions by monetary and fiscal policy, as well as the now added change of inflation targeting.

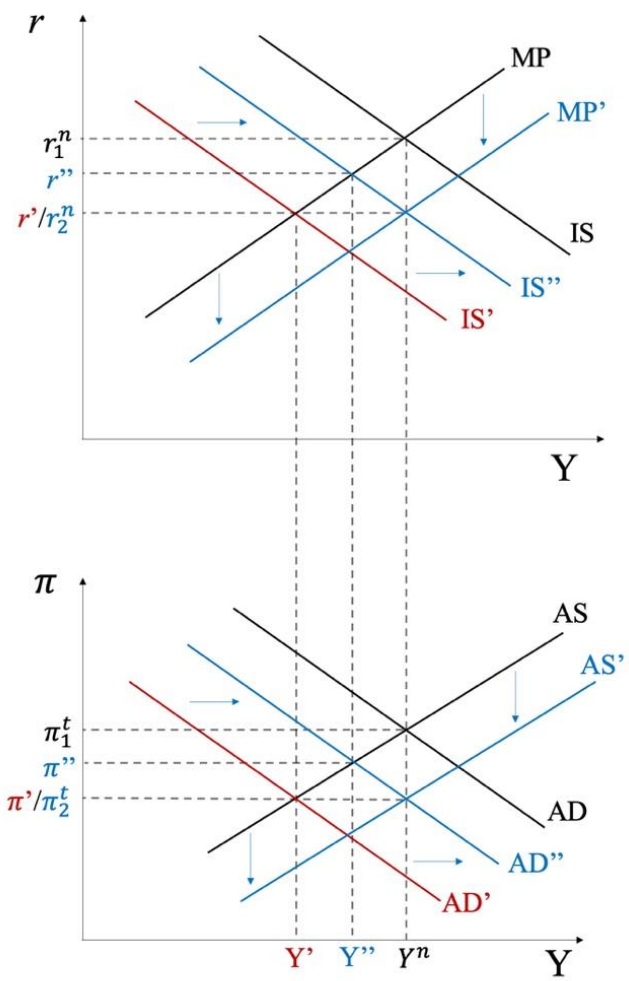

Figure 8: Using monetary policy to counteract the negative demand shock in IS/MP-AD/AS model. Adapted from Lukkezen J., Jacobs B. & Kool C. (2016) by authors.

3.2 Future Outlook

The previous analysis assumed that the shock experienced was only temporary. However, many indicators point to a reality of continuously diminishing oil returns with persistent economic effect on Norway. The OECD (2018) claim in their survey on Norway that the government needs to prepare its economy for this approaching development and recommends multiple approaches and policies that all aim to strengthen other industries and businesses. One suggestion in the Economic Outlook of the OECD (2019) states that a reduction of the corporate and income tax rate would benefit Norwegian companies by being in a more competitive environment. Furthermore, a tax rate reduction would allow business to increase their investment due to higher financial resources and increase individual's disposable income, which could lead to higher consumption. In fact, Norway lowered its corporate and basic income tax rate in the past years (Norwegian Tax Administration, 2019). Moreover, the social welfare fund allows Norway to reduce their tax rates without necessarily causing their budget deficit to increase (see Appendix Figure 13). However, diminishing returns from the petroleum sector also result in lower returns on the social welfare fund. For that reason, OECD (2019) emphasise the importance of efficient investment by the Norwegian government today, in order to support diversification of the economy.

Continuing with the assumption that diminishing oil returns and reserves have a permanent adverse shock on the economy, a lower natural rate of interest is necessary to sustain potential output. Through the lowering of the inflation target in 2018, this phenomenon is actually visible. Figure 8 illustrates how the same potential output is now achieved by a lower natural rate of interest implying permanent tendencies for the shock. Furthermore, depleting natural resources imply tighter productivity-constraints for an economy in the future, thus possibly lowering its potential output (OECD, 2019). Since fiscal and monetary expansions do not provide feasible growth above potential output in the long run, Norway will need to perform structural reform measures as mentioned by Lukkezen, Jacobs and Kool (2016). Structural reforms increase the potential output of an economy and entail policies such as the abovementioned lower tax burdens, as well as public investment in areas such as education, infrastructure and research. Within the IS/MP-AD/AS model as it was used in this paper, those measures would allow for sustainable growth in the long-run by counteracting the decrease in natural caused by limited natural resources. The policy measures recommended by the OECD to sustain economic growth conform with the structural reform measures required by the model. Lower tax burden allows businesses to more easily compete on the worldwide market, while increasing demand and

investment. Public investment to prepare Norway's economy for a continuously shrinking petroleum sector should be focused in infrastructure and education to sustain a high level of potential output.

4. Conclusion

Norway finds itself in a unique position. It possesses an economy that was largely dominated by its petroleum sector in the last decades, but also a high amount of financial wealth through clever investment of its petroleum revenue. Despite lower oil production and a dwindling petroleum sector in recent years (see Appendix Figure 9 and 12), Norway's whole economy still heavily depends on its offshore industry. This paper illustrated this dependency by first observing exports and exchange rate and correlating them to worldwide oil prices. Further, it focused its analysis on the recent price drop of oil in 2014 and inspected the responses of Norwegian institutions by applying a simple IS/MP-AD/AS model and looking at empirical data. The model and data of that time period suggest that the negative output gap was overcome in 2018, however with lower inflation and natural rate of interest, due to monetary policy changing its inflation target in 2018. Lastly, the paper linked its model to the OECD Economic Outlook of 2019 to compute a future outlook for the Norwegian economy. It concludes that the general development of diminishing returns in the petroleum sector can be considered a permanent adverse shock, especially given the scarcity of natural resources. Structural reform measures would be able to offset this negative pressure and sustain economic growth for Norway in the long run. Those measures include lower tax burdens as well as investment in the infrastructure and education of Norway, which go accord with the recommendations stated by the OECD.

5. Appendix

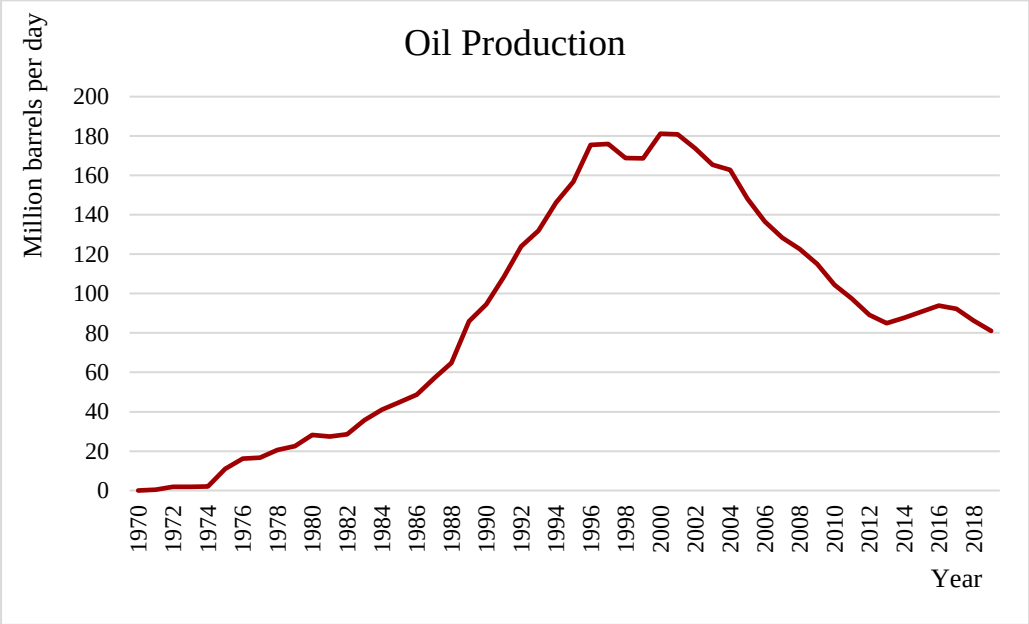

Figure 9:Oil production in million barrels per day. Retrieved from Ministry of Petroleum and Energy (2019).

Figure 9 illustrates the oil production from 1970 to 2018 in million barrels per day. It is evident that after finding the petroleum offshore, Norway started to produce oil increasing its production until the early 2000s. From 2002 onwards, production decreased due to the limited resources.

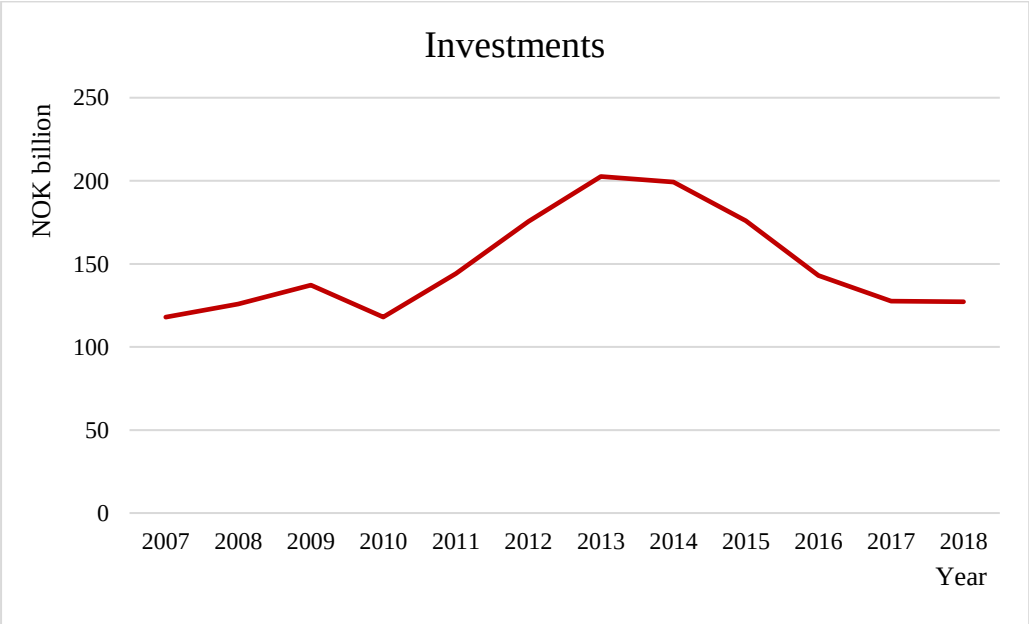

Figure 10: Investments in petroleum activities. Retrieved from Ministry of Petroleum and Energy (2019).

Figure 10 shows the investment in the petroleum activities in NOK billion. As the price of oil dropped in 2014, Norway's incentives to invest in the petroleum activities decreased as well.

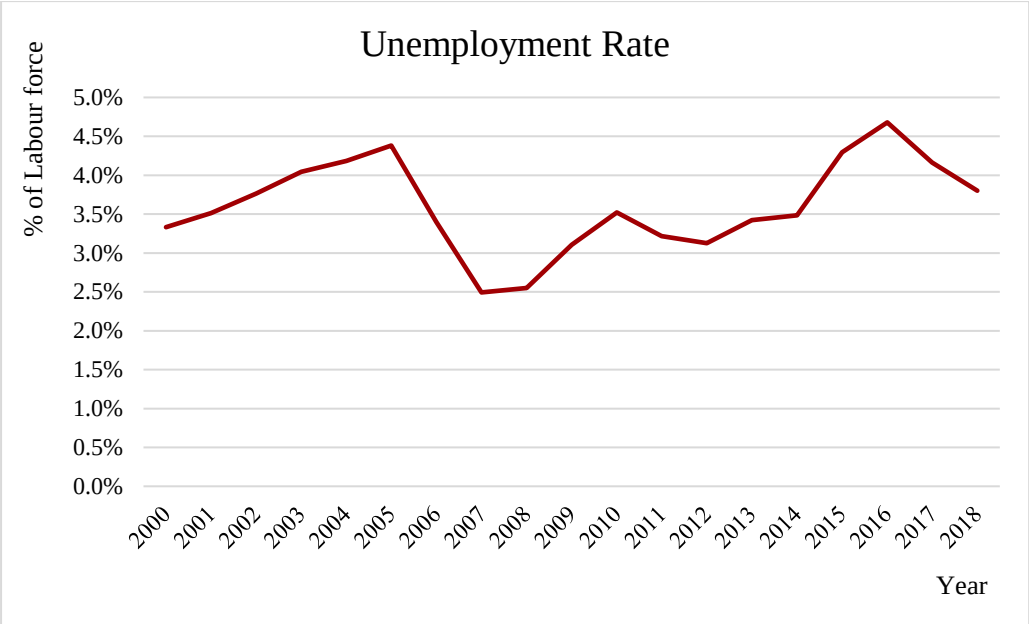

Figure 11: Unemployment rate. Retrieved from OECD (2019).

Figure 11 illustrates the unemployment rate in percentage of the labour force. During price drops in oil price, Norway has less incentives to extract the oil offshore because it is highly costly. Therefore, as the oil price decreases in 2014, less workers are needed and therefore unemployment rate increases.

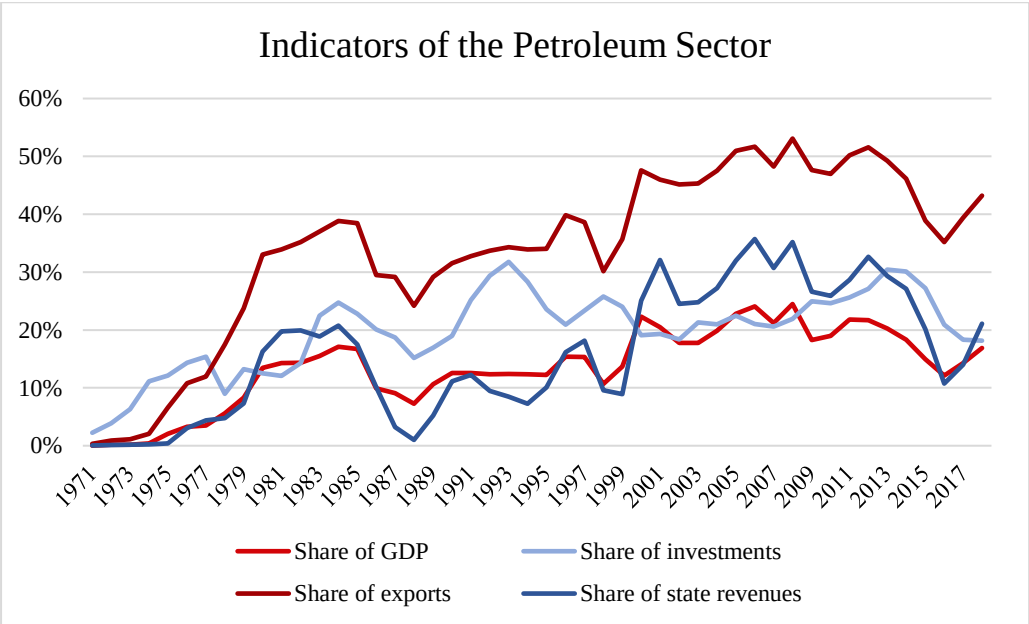

Figure 12: Indicators of the petroleum sector. Retrieved from Ministry of Petroleum and Energy (2019).

Figure 12 displays the share of petroleum related activities in different macroeconomic indicators. This Figure shows how much petroleum activities remain important to Norway's economy, even fifty years after the discovery of oil.

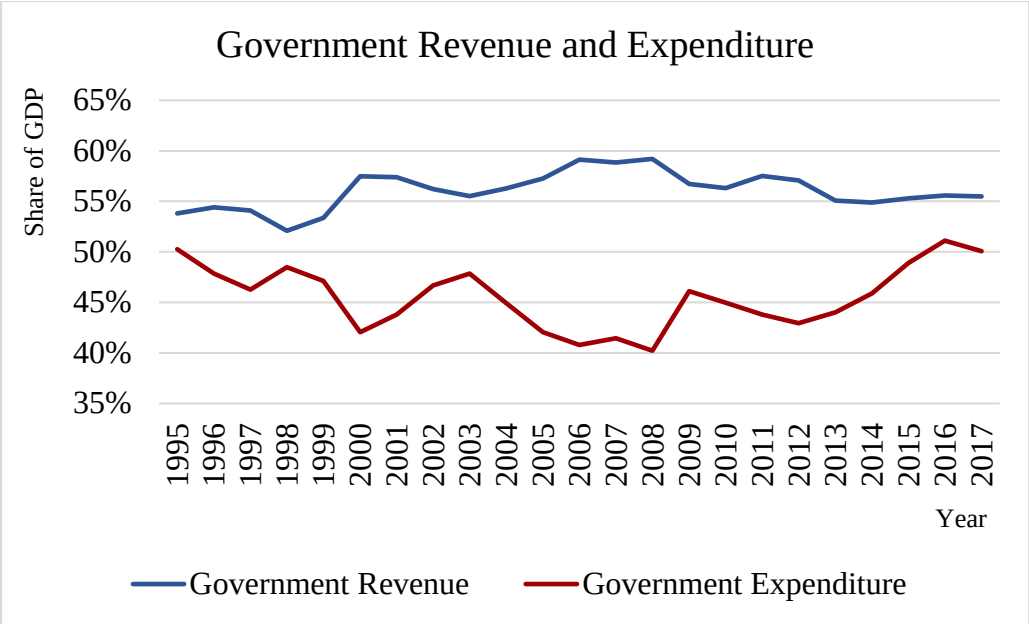

Figure 13: Government revenue and government expenditure. Retrieved from OECD (2019).

Figure 13 depicts the government revenue, in blue, and the government expenditure, in red, as a percentage share of GDP. It is clearly shown, in Figure 13, that the government has a budget surplus from 1995 onwards, especially due to the government fund.

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